



College of Science & Mathematics

BIOL 7950: Directed Study

Course Section: 01 (88743)

Fall Semester 2026 Syllabus

Instructor Information

Nicholas Green

Email: ngreen62@kennesaw.edu

Office Location: SC 331

Office Hours: By appointment

Preferred Method of Communication: Email

Course Description

Course Description: The course content is a concentrated investigation of selected, advanced topics, which may include original research projects. The course content will be determined jointly by the instructor and the student.

Course Information

Total Credit Hours: Variable

Laboratory / Experiential Field Hours: Variable

Prerequisite: None

Course Meeting Information

Course Meeting Time: Tuesdays 12 - 1 pm

Course Meeting Location: TBD

Course Meeting Modality: In person

Course Materials

None

- There are no course materials required for this course.

Course Learning Outcomes

- Analyze data using Bayesian inference in the JAGS software package.
- Communicate the results of Bayesian data analyses to audiences who may or may not have statistical backgrounds.
- Explain Bayes' theorem and how it relates to statistical inference.
- Explain the differences between, and relative advantages of, Bayesian and frequentist inference.

Teaching Methods

This course will use the following teaching methods:

- Common Intellectual Experiences
- Hand-on Learning Experiences / Laboratory Experiments
- Lecture

Course Requirements and Assignments

- **Presentations:** Each student will be expected to present a Bayesian data analysis to the class. This can come from their own work or from a published paper. Students will be evaluated based on how well they present, explain, and answer questions about the analysis. A detailed assignment guide will be provided early in the semester.
- **Midterm and final exams:** These exams are used to evaluate student understanding and progress at the middle and end of the semester. Exams are take-home, open book, and open note. Collaboration between students is encouraged.

Evaluation and Grading Information

The presentation, midterm, and final exam each constitute one-third of the course grade. I.e., the course grade (percentage) is the mean percentage score on the presentation, midterm, and final exam.

Feedback on all assignments will be returned within 7 calendar days of when the assignment was turned in.

There is no extra credit available in the course.

Course letter grades are assigned according to the traditional 10% scale ($\geq 90\%$ = A, 80-89% = B, etc.). Percentages are rounded to the nearest integer.

Course Calendar

Week	Date	Topic	Assignment(s)
1	Aug 25	No class	
2	Sept 1	Bayes theorem and Bayesian inference	
3	Sept 8	Introduction to JAGS	
4	Sept 15	The Bayesian workflow in R	
5	Sept 22	Linear models in JAGS part 1	
6	Sept 29	Linear models in JAGS part 2	
7	Oct 6	Bayesian GLM part 1	Midterm exam due Sunday Oct 11 @ 11:59 pm
8	Oct 13	Bayesian GLM part 2	
9	Oct 20	Nonlinear models	
10	Oct 27	Mixed models part 1	
11	Nov 3	Mixed models part 2	
12	Nov 10	Mark-recapture models	
13	Nov 17	Student presentations	
14	Nov 24	Thanksgiving break - no class	
15	Dec 1	Student presentations	Final exam due Sunday Dec 6 @ 11:59 pm

Course Policies

- Attendance is required and taken at each course meeting. The maximum possible letter grade a student can earn is determined by the number of unexcused absences: 0 to 2 for an A, 3 to 4 for a B, 5 or 6 for a C, 7 to 8 for a D, and more than 8 for an F.*
- Absences can be excused for reasons such as serious illness of self or a loved one, death in the family, jury duty, military service, or other reasons at the discretion of the instructor.*

- *Deadlines for assignments can be extended at the discretion of the instructor.*
- *Collegial and respectful behavior towards all people is expected at all times.*
- In this class, you are welcome to use generative “artificial intelligence” (hereafter “AI”) tools for any purpose other than taking exams. Generative AI is here to stay, so it is incumbent on all of us to learn to use it responsibly. Here’s the important thing to remember: AI should be a supplement to your knowledge and skills, not a replacement for knowledge and skills.
- You should note that all AI generative tools tend to make up incorrect facts and fake citations, code generation models sometimes to produce inaccurate outputs, and image/art generation tools can produce copied work or offensive products. You will be responsible for any inaccurate, biased, offensive, or otherwise unethical content you submit regardless of how it was created.
- If you use an AI tool on an assignment or exam, you must explicitly describe its contribution in your submission. This includes a PDF of the entire transcript, including prompts and outputs. The use of an AI tool without acknowledgement is cheating and constitutes a violation of the KSU Code of Academic Integrity.
- The application of statistical methods in biological investigations can be highly nuanced and context-specific, and AI may not understand every situation accurately. Never accept an AI’s output at face value, and check what it says against other sources or knowledgeable colleagues.
- If you use AI to help you generate R code, run and check the code yourself to verify that (1) the code works; and (2) your instructions were followed accurately. Just as a human colleague could misunderstand your instructions and write unhelpful code, so could an AI.

Modified Operations Policy:

Kennesaw State University is committed to remaining open and continuing full operations, including class schedules, whenever possible. The Office of the President, in consultation with the KSU Office of Emergency Management and other officials, will determine when an emergency impacts KSU’s normal operations. When the University is closed due to an emergency, all classes will be canceled, and non-essential personnel will not be expected to work.

When the University shifts to modified operations, Cabinet members, in consultation with the Office of the President or designee and the Office of Emergency Management, will be responsible for determining appropriate

operational continuity for their divisions. This may include remote work. During modified operations, each faculty member will be responsible for instructional continuity for their classes. To ensure instructional continuity, all Academic Colleges and Departments are expected to maintain and regularly update instructional continuity plans. All syllabi should also include an instructional continuity plan statement specific to that course.

During a weather event or other emergency, those commuting to campus should exercise their best judgment when determining travel options. If the University remains open and an employee elects not to report to work, the employee will be required to use their accrued vacation time.

The University will announce campus closures and delayed schedules in several ways. The cell phone number on file with the university will automatically receive KSU Alerts, so make sure your information in OwlExpress is accurate at all times. An email will also be sent to your university account.

In addition, announcements will be made by a notice on the Kennesaw State University home page.

Course continuity plan

- **Communication:** Please check D2l Brightspace or e-mail for necessary instructions.
- **Virtual Classes:** If in-person classes are not possible, we may transition to virtual classes using MS Teams.
- **Assignments and Assessments:** Deadlines for assignments and assessments may be adjusted to accommodate the emergency situation.

Institutional Policies

Academic Integrity Statement:

Every KSU student is responsible for upholding the provisions of the [Student Code of Conduct](#), as published in the Undergraduate and Graduate Catalogs. Section 5c of the Student Code of Conduct addresses the university's policy on academic honesty, including provisions regarding plagiarism and cheating, unauthorized access to university materials, misrepresentation/falsification of university records or academic work, malicious removal, retention, or destruction

of library materials, malicious/intentional misuse of computer facilities and/or services, and misuse of student identification cards. Incidents of alleged academic misconduct will be handled through the established procedures of the Department of Student Conduct and Academic Integrity (SCAI), which includes either an “informal” resolution by a faculty member, resulting in a grade adjustment, or a formal hearing procedure, which may subject a student to the Code of Conduct’s minimum one semester suspension requirement.

KSU Disruption of Campus Life policy:

All students are responsible for knowing the information, policies and procedures outlined in the Kennesaw State University Codes of Conduct. [Student Conduct and Academic Integrity \(SCAI\)](#) includes: the general Student Code of Conduct, the Residential Code of Conduct, and the Code of Academic Integrity.

KSU Web Accessibility:

Federal law Section 508 Subsection 1194.22 of the Rehabilitation Act and the Board of Regents (BOR) of the University System of Georgia (USG) Web Accessibility Guidelines require that all web content meet the federal government’s accessibility guidelines. As such, KSU complies with USG guidelines.

University accessibility assistance is provided by several offices as noted below. Staff in these offices work to accommodate requests for access or assistance with access as soon as possible in order to either accommodate the request or identify an effective alternative for the requester.

Request for Office Contact Number Contact Email

- Student Support Services Student Disability Services 470-578-2666 sds@kennesaw.edu
- Student Technology Assistance [Student Technology Assistance](#)
- University Information Technology Services 470-578-6999 service@kennesaw.edu
- Third Party Technology Assistance University Information Technology Services 470-578-6999 service@kennesaw.edu

KSU Reasonable Accommodations:

Students with qualifying disabilities under the Americans with Disabilities Act (ADA) and/or Section 504 of the Rehabilitation Act who require “reasonable accommodation(s)” to complete the course may request those from Office of Student Disability Services. Students requiring such accommodations are required to work with the University’s Office of Student Disability Services rather than engaging in this discussion with individual faculty members or academic departments. If, after reviewing the course syllabus, a student anticipates or should have anticipated a need for accommodation, he or she must submit documentation requesting an accommodation and permitting time for a determination prior to submitting assignments or taking course quizzes or exams. Students may not request retroactive accommodation for needs that were or should have been foreseeable. Students should contact the office as soon as possible in the term for which they are seeking accommodations. Student Disability Services is located in Kennesaw Hall, Room 1205 on the Kennesaw campus and Building A in Suite 160G on the Marietta campus. Please visit the [Student Disability Services \(SDS\) website](#) for more information, or call the office at 470-578-2666 (Kennesaw campus) or 470-578-7361 (Marietta campus).

KSU Enrollment Management/Course Attendance:

Students are solely responsible for managing their enrollment status in a class; nonattendance does not constitute a withdrawal.

Copyright Law:

It is the responsibility of KSU faculty and students to respect the rights of copyright holders and complying with copyright law. The University System of Georgia recognizes that the exclusive rights of copyright holders are balanced by limitations on those rights under federal copyright law, including the right to make a fair use of copyrighted materials and the right to perform or display works in the course of face-to-face teaching activities.

The University System of Georgia facilitates compliance with copyright law and, where appropriate, the exercise in good faith of full fair use rights by faculty and

staff in teaching, research, and service activities. The University System of Georgia ensure compliance with copyright law in the following ways.

1. The USG informs and educates students, faculty, and staff about copyright law, including the limited exclusive rights of copyright holders as set forth in 17 U.S.C. § 106, the application of the four fair use factors in 17 U.S.C. § 107, and other copyright exceptions.
2. The USG develops and makes available tools and resources for faculty and staff to assist in determining copyright status and ownership and determining whether use of a work in a specific situation would be a fair use and, therefore, not an infringement under copyright law;
3. The USG facilitates use of materials currently licensed by the University System of Georgia and provides information on licensing of third-party materials by the University System; and
4. The USG identifies individuals at the University System and member institutions who can counsel faculty and staff regarding application of copyright law.

Protecting Students' Privacy (FERPA):

Students have certain rights to privacy. These rights are mandated by federal policy. [The Student Handbook](#) contains information regarding Rights Pertaining to Student Records, and FERPA specific details are available on the [KSU FERPA website](#).

A key requirement of the formal evaluation process is the protection of individual privacy rights concerning educational grading. The University's online learning system and email system is designed to prevent unauthorized individuals from gaining access to sensitive information or information protected by federal or state law. Consequently, faculty and students are strongly encouraged to only communicate regarding course matters through the University's designated technology learning system.

Information should not be made public in any way in which a student's grades, social security number, or other personal information may be identified. Grade information may be shared with members of the KSU community who also have a legitimate educational interest in student success (e.g. academic advisors or members of the Behavioral Response Team). Faculty may be asked to provide early alert information if there is a concern that a student is at risk, academically or otherwise.

KSU Sexual Misconduct:

In accordance with federal and state law including, Title IX of the Education Amendments of 1972 ("Title IX") and Title VII of the Civil Rights Act of 1964 (Title VII), the University System of Georgia (USG), including Kennesaw State University, prohibits discrimination on the basis of sex in any of its education programs or activities or in employment. The USG is committed to ensuring the highest ethical conduct of the members of its community by promoting a safe learning and working environment. To that end, Kennesaw State University follows USG Board of Regents Policy Manual, Section 6.7. [Visit Title IX](#).

Withdrawal from Classes:

A student who officially withdraws from a course by the end of the last day to withdraw without academic penalty will receive a grade of "W" and receive no credit.

A student will receive a refund only when the student withdraws from ALL courses for the applicable semester and only by the schedule outlined in the University refund policy found in the Tuition, Expenses, & Financial Aid section of the undergraduate and graduate university catalogs.

Students should be aware that a reduction in their hours might result in the loss of full-time student status and thus affect their financial aid, scholarships, athletic and ticket eligibility, University housing accommodations, use of University resources and access to University facilities, visa for international students, and Veterans Educational Benefits. Students should contact the appropriate office and their academic advisor with questions about the impact of their withdrawal from a course before initiating a withdrawal. Veterans and dependents of veterans who receive educational benefits must notify the Veterans Education Benefits Area in the Office of the Registrar of any course load reductions.

Hardship Withdrawals

If a student experiences significant personal hardship (e.g., medical or family emergency, prolonged illness), the Dean of Students can approve a hardship withdrawal from all courses in the term for which the student is currently

registered. In the case of an approved hardship withdrawal from all courses, the Registrar will assign grades of “W” for those classes. The deadline for final approval of a hardship withdrawal by the Dean of Students is the last day of class for which the hardship withdrawal is sought. If the hardship withdrawal process is not complete by the last day of class for which the hardship withdrawal is sought, a student must appeal for a retroactive hardship withdrawal from the Academic Standing Committee.

Appeals for retroactive hardship withdrawals must be directed to the Academic Standing Committee. Retroactive hardship withdrawals are rarely granted if it has been more than one year since the last day of class for which the withdrawal is sought. Extraordinary justification must be shown. In the case of approved retroactive hardship withdrawals, the Registrar will assign a grade of “W.”

University Initiated Withdrawals

If a student is suspended by the Office of Student Conduct following a violation of the University’s Code of Conduct not related to academic dishonesty, the Office of Student Conduct may facilitate a University-initiated withdrawal from courses for which a student is registered for the term. The Registrar will assign grades of “W” for those classes.

Military Withdrawals

A student will receive a “WM” symbol for all courses and a full refund of tuition and mandatory fees and a pro rata refund of other fees for military and other service, as defined by [BOR Policy Manual, Section 7.3.5.3](#). To request a military withdrawal, the student must submit a copy of official orders to the Office of the Registrar.

Academic Feedback:

Institutional Chief Academic Officers will encourage faculty to clarify for students, at the beginning of each course, the basis on which grades will be determined and to provide timely academic feedback as the course progresses ([BOR Academic and Student Handbook Policy 2.18](#))

Netiquette: Communication Courtesy:

All members of the class are expected to follow rules of common courtesy in all email messages, threaded discussions and chats. See [Email Best Practices](#) for additional guidance.

Course Delivery:

KSU may shift the method of course delivery at any time during the semester in compliance with University System of Georgia health and safety guidelines. In this case, alternate teaching modalities that may be adopted include hyflex, hybrid, synchronous online, or asynchronous online instruction. Please refer to the Instructional Continuity statement included in each course syllabus for specific plans.

University Sponsored Academic Resources

Academic resources supporting KSU's current graduate students can be found here: <https://www.kennesaw.edu/graduate/current-students/index.php>

Additional academic resources include

- [Disability Resources](#)

Technology Resources

For issues with technical difficulties, please contact the Student Help Desk:

[Submit a Request for Help](#)

Email: service@kennesaw.edu

Call: 470-578-6999

[New Students Technology Services](#)- Steps that will help you meet the technological requirements of the University.

Any request for make-ups due to technical issues **MUST** be accompanied by the

ticket number received from UITs when the problem was reported to them. The ticket number will document the time and date of the problem. You MUST e-mail your instructor within 24 hours of the technical difficulty if you wish to request a make-up.

Student Support and Wellness Resources

Student Support Resources

- [Department of Career Planning & Development](#)
- [Student Health Clinic](#)
- [Military and Veteran Services](#)

Student Wellness Resources

YOU MATTER at KSU, and your wellbeing is a priority to us. College is exciting, but it can also bring challenges that affect your health, wellbeing, and academic success. Whether you need academic accommodations through Student Disability Services, counseling, health education, addiction and recovery support, fitness and recreation opportunities, or access to basic needs resources, [Wellbeing@KSU](#) is here to help you do well and be well.

Faculty Profile (USG CV Requirement):

Directory Information

Name: Nicholas Green

Title: Associate Professor of Biology

Email: ngreen62@kennesaw.edu

Home department: SCM- Ecology, Evolution, and Organismal Biology

Academic department: SCM-Ecol, Evol, & Organism Biol

Phone: 470-578-6546

Location: Kennesaw Campus > 370 Science > Floor 3 > 331

Mail Drop: 1202

Degrees Earned

BS in Biology, University of Louisville

PhD in Biology, Baylor University

Teaching Interests

My teaching is concentrated in two areas: vertebrate biology, and biological data analysis. One of my primary goals for all of my courses is to improve the quantitative and statistical reasoning skills of students. This often means introducing students to state-of-the-art methods in data collection and analysis, often far beyond what most biology majors might otherwise encounter in their undergraduate careers.

Research Interests

1. The [Green Quantitative Ecology Lab \(GreenQuantEco\)](#) uses advanced mathematical and statistical tools to investigate relationships between living organisms and their environment. Most of our research focuses on understanding how human activities, like urbanization and climate change, affect fish and wildlife across all levels of biological organization.
- 2.
3. While our research is often at the cutting-edge of quantitative methods, we are grounded in a rigorous tradition of field ecology and mammalogy. Our work takes us all over Georgia, from the northern Ridge and Valley to the barrier islands of the coast. To complement our computational and field work, we use a wide variety of lab methods like DNA extraction, ELISA, necropsy, craniometry, and microscopy gather the data we need from the samples we collect.

Recent Scholarship

A complete list of publications can be found at [this link](#).